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High Performance CO_2 Foam Fracturing Fluid with Surfactant and Crosslinked Associative Polymer Gel for High Temperature Reservoir

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Abstract

Objective/Scope: CO_2 foam fracturing is recognized as a technology for reducing carbon emissions and conserving water. Compared to conventional water-based fluids, it offers several advantages including improved flowback rates, minimized formation damage, high proppant pack conductivity, and stimulation of water-sensitive formations, benefiting hydrocarbon production improvement and cost reduction. However, current CO_2 foam fracturing technology is limited to reservoirs below 120°C. This study presents a new CO_2 foam fluid with temperature resistance up to 150°C.

Surfactants, associative polymers and crosslinkers were screened and optimized to develop a surfactant with crosslinked associated polymer formulation (SCAP). A CO_2 foam fracturing fluid was then developed based on SCAP gel. The foam fluid formulation was further optimized with other additives including clay stabilizer, flowback enhancer, and capsule breaker to achieve a CO_2 foam fracturing fluid package. The performances of optimized CO_2 foam fracturing fluid were investigated including foam stability, proppant suspension capacity, rheological properties and breakability respectively.

A SCAP fracturing fluid was developed using 0.5% surfactant, 0.5% associated polymer and 0.5% organic crosslinker, exhibiting thermal thickening behavior and preserving viscosity above 300 mPa.s at shear rate of 100 s^{-1} over 2 hours at temperatures ranged from 90°C to 150°C. Moreover, the interaction between surfactant and crosslinked gel was investigated and confirmed for enhanced viscosity at elevated temperature. Leveraging this development, a CO_2 foam fracturing fluid was further formulated using SCAP gel. The formulation was optimized with foam quality, polymer addition and thermal stabilizer to ensure its high performance at 150°C. The optimized package includes 0.5% surfactant, 0.8% associative polymer, 0.5% crosslinker, and 0.5% thermal stabilizer. The CO_2 foam with 60% foam quality maintains high viscosity above 300 mPa.s at shear rate of 100 s-1 for over 2 hours with temperature up to 150°C. Foam stability tests and static proppant suspending experiments demonstrates the developed CO_2 foam fracturing fluid exhibits excellent foam stability and no obvious proppant setting at 150°C during 3 hrs tests. After a static break test with the addition of 0.02% capsule breaker, the foam fracturing fluid fully degrades into

a watery fluid, indicating low formation damage. The coreflooding tests further confirmed low formation damage of the fluid, with retained permeability reaching approximately 87.5%.

This work significantly extends the operational envelope of CO_2 foam fracturing to high temperature conditions, benefiting hydrocarbon production and operational efficiency, while effectively enriching CO_2 utilization and reducing carbon footprint.

Introduction

Hydraulic fracturing has become a leading stimulation technique for improving oil and gas production, especially for unconventional oil and gas resources such as shale formations and tight sands. The fracturing process involves injecting well-formulated fracturing fluid into subsurface formation under high pressure to create artificial fractures and propagate fracture networks. These fractures serve as highly conductive pathways that substantially improved the permeability of the target reservoir. Proppants typically sand and ceramic particles are pumped with the fluid and left within the fractures to keep them open once completing the fracturing process. The resulting fracture networks improve hydrocarbon flow, thereby benefiting economical hydrocarbons production (Vidic et al. 2013).

Fracturing fluids as a critical component plays an important role in transferring energy from pumps to create fractures, which have been extensively developed to meet the increasing demands from reservoir stimulation in the past decade (Gandossi 2013). Generally, water-based fluids are extensively utilized to induce fracture due to their cost-effectiveness, and ease of field operation(Lei et al. 2021). However, water-based fluids encounter several challenges in field operation, including formation water sensitivity, water blocking, water availability and formation clean up for low pressure reservoirs (Ahmed et al. 2019, Li et al. 2022). Besides, utilization of water-based fluids requires large volume of water, which not only raises environmental concerns related to water resource conservation and wastewater management, but also imposes considerable logistical challenges, particularly in remote or arid regions where water resources are limited (Almubarak et al. 2020). These increasing concerns over environmental protection, water conservation, and formation damage have accelerated the demand for development of alternative fluids. Among these, carbon dioxide (CO_2) foam fracturing fluids using a small amount of water have developed as a promising alternative to water-based ones, offering a unique combination of environmental and operational advantages.

CO_2 foam fracturing fluids has been recognized as a promising fracturing fluid since 1980s (Wamock et al. 1985, Garbis and Taylor 1986, Reidenbach et al. 1986, Phillips et al. 1987). It has been demonstrated that CO_2 foam fracturing fluid can significantly improve hydrocarbon production and reduce cost compared to water-based fracturing fluids (Friehauf et al. 2009, Burke and Nevison 2011, Denney 2010). CO_2 Foam fracturing fluid displays advantages including enhancing flowback rate of fracturing fluid with minimized formation damage, retaining high proppant pack conductivity, stimulating water-sensitive formation, saving water resource and reducing carbon footprint (Wamock et al. 1985, Harris 1992, Wilk et al. 2015, Nianyin, Chao, et al. 2021, Mojid et al. 2021, Nianyin, Jiajie, et al. 2021) (Fig. 1). Moreover, injected CO_2 can enhance natural gas production in shale reservoirs through competitive adsorption, where CO_2 displace adsorbed methane (CH_4) from the shale surface. This is due to the strong affinity of CO_2 in the shale surface, making it effectively displace methane.

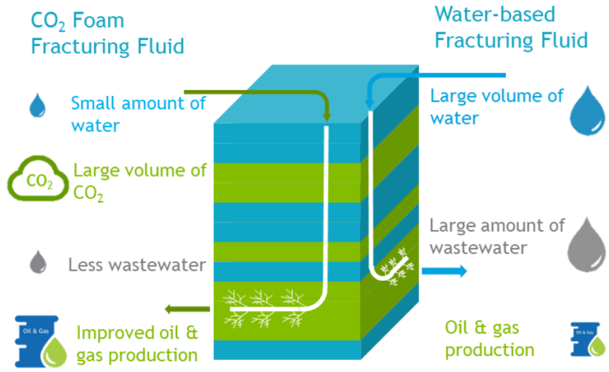

Figure 1—Comparison of CO_2 foam fracturing fluid with water-based fracturing fluid

Despite these advantages, the application of CO_2 foam fracturing fluids remains limited by several technical challenges. One key issue is the high diffusivity of CO_2 molecule that can easily penetrate the chemical layer structures, thereby compromising foam stability, particularly when compared with N_2 molecules (Zhao et al. 2024). This instability becomes more pronounced at elevated temperatures, where CO_2 foam tends to collapse, leading to poor proppant transport and ineffective stimulation. Besides, CO_2 dissolved in aqueous can generate carbonic acid, lowering the pH to around 3~5. This acidic environment requires CO_2 foaming agents, viscosifiers and other additives that are compatible under such conditions. However, it is very challenging to ensure the compatibility of chemicals with acidic environment, especially for the crosslinked polymer gels. CO_2 foam systems often rely on crosslinked polymer gels like borate crosslinked guar gum and Zirconium crosslinked CMHPG systems, to increase fluid viscosity and ensure good foam stability. Unfortunately, these crosslinkers are incompatible with the acidic environment, causing premature break or failure to crosslink in high pH gel systems (Harris 1992, Samuel et al. 2023). Moreover, acidic environment can catalyze the breakage of the crosslink bond and accelerate polymer degradation through hydrolysis, further reducing its temperature resistance(Zhao et al. 2024). To address these limitations, extensive research has focused on improving the performance of chemicals in acidic condition. Strategies include guar gum molecular modification(Qiu et al. 2018, Wang et al. 2012), development of new synthetic polymers (Eoff et al. 2005, Xu et al. 2019, Xu 2018, Qiao 2021), design of new acid tolerant crosslinkers (Sokhanvarian et al. 2019, Zhang et al. 2019) and addition of nanoparticles(Emrani and Nasr-El-Din 2017, Thakore et al. 2020, Fu and Liu 2021). In parallel, viscoelastic surfactant (VES) (Xue et al. 2016, Feng et al. 2022, Zheng, Zhu, Yang, Jiang, et al. 2025, Zheng, Zhu, Yang, Song, et al. 2025) and non-crosslinked polymer like associative polymers(Xiong et al. 2021, Sun et al. 2023, Samuel et al. 2023, Gao et al. 2024, Beletskaya et al. 2025) have been explored as new alternatives. These chemicals can provide good proppant suspending capacity for fracturing operation with low viscosity solution, due to their excellent elastic properties. However, their application is constrained by high costs due to the need for the high additions and limited temperature resistance as elevated temperatures accelerate molecular motion and weaken the network structure (Gupta et al. 2005, Gupta 2009, Gupta and Hlidek 2009). As a result, these constraints have restricted the operational envelope of CO_2 foam fracturing fluids, confining their use to reservoirs with temperatures below 120°C (Gupta and Tudor 2005, Gupta et al. 2005, Gupta and Hlidek 2009, Yekeen et al. 2018).

To address the limitations of current CO_2 foam systems and unlock their full potential in high-temperature and tight formations, this study focuses on development of high-performance CO_2 foam fluids capable of operating at temperatures up to 150°C. Surfactants, associative polymers and crosslinkers were screened and optimized to develop a surfactant-crosslinked associated polymer (SCAP) formulation. A CO_2 foam fracturing fluid was then developed based on SCAP gel. The foam fluid formulation was further optimized with integration of other additives including clay stabilizer, flowback enhancer, and capsule breaker to achieve a comprehensive CO_2 foam fracturing fluid package. The performances of the fluid package including foam stability, proppant suspending capacity, rheological properties, controlled breakability and formation damage were thoroughly evaluated at 150°C.

Experimental

Materials

Chemicals. Table 1 summarizes the information of chemicals used in this study, including surfactants, associated polymers, crosslinkers, thermal stabilizer, clay stabilizer, breaker and flowback enhancer in this work. All chemicals were used as received, without further purification.

Table 1—Information of surfactants

No.	Sample	Type	Apparent Form
1	Surf-1	Zwitterionic surfactant	Liquid
2	Surf-2	Zwitterionic surfactant	Liquid
3	Surf-3	Zwitterionic surfactant	Liquid
4	Surf-4	Cationic surfactant	Powder
5	Surf-5	Anionic surfactant	Powder
6	APAM-1	Associated polymer	Powder
7	APAM-2	Associated polymer	Powder
8	APAM-3	Associated polymer	Powder
9	APAM-4	Associated polymer	Powder
10	APAM-5	Associated polymer	Powder
11	APAM-6	Associated polymer	Powder
12	APAM-7	Associated polymer	Powder
13	APAM-8	Associated polymer	Powder
14	APAM-9	Associated polymer	Emulsion with 32% content
15	APAM-10	Associated polymer	Emulsion with 32% content
16	APAM-11	Associated polymer	Powder
17	APAM-12	Associated polymer	Powder
18	OCX-1	Organic Zirconium crosslinker	Liquid
19	OCX-2	Organic Zirconium crosslinker	Liquid
20	B-Cap	Capsule breaker	Powder
21	TU	Thermal stabilizer	Powder
22	ClayS-1	Clay stabilizer	Liquid

Table 1a—Information of surfactants

No.	Sample	Type	Apparent Form
1	Surf-1	Zwitterionic surfactant	Liquid
2	Surf-2		Liquid
3	Surf-3		Liquid
4	Surf-4	Cationic surfactant	Powder
5	Surf-5	Anionic surfactant	Powder
6	APAM-1	Associated polymer	Powder
7	APAM-2		Powder
8	APAM-3		Powder
9	APAM-4		Powder
10	APAM-5		Powder
11	APAM-6		Powder
12	APAM-7		Powder
13	APAM-8		Powder
14	APAM-9		Emulsion with 32% content
15	APAM-10		Emulsion with 32% content
16	APAM-11		Powder
17	APAM-12		Powder
18	OCX-1	Organic Zirconium crosslinker	Liquid
19	OCX-2		Liquid
20	B-Cap	Capsule breaker	Powder
21	TS-1	Thermal stabilizer	Powder
22	ClayS-1	Clay stabilizer	Liquid

Water. Tap water, whose composition is detailed in Table 2, was used for fluid preparation.

Table 2—Composition of tap water

Ion	Concentration, ppm
Na^+	21
Ca^{2+}	10
Mg^{2+}	19
Cl^-	105
TDS	115

Polymer Fracturing Fluid Preparation

500 ml tapwater was introduced into a flask and stirred at 500 rpm by a magnetic stir to obtain a good vortex. A calculated polymer powder was then gradually added over into the solution in 5 minutes to ensure uniform dispersion. Following the addition, the stirring speed was reduced to 200 rpm to avoid mechanical degradation of polymer. Subsequently, the solution was kept at room temperature without stirring for another 24 h to assure proper hydration. After that, a calculated amount of crosslinker was added to the solution, resulting in the formation of a high-viscosity gel.

Gelation Time Measurement

The prepared polymer solution was stirring to estabilish a vortex. Then a calculated amount of crosslinker was added to the solution. Stirring was continued until the vortex disappeared, at which point it was stopped. The time required for vortex close was recorded as gelation time.

CO_2 Foam Fluid Preparation

A measured volume of prepared solution was transferred into flask. The solution was mixed with injecting CO_2 gas at 10000 rpm/min by high-speed mixer within 2 mins. The crosslinker then was added to the mixture with stirring in 1 minute to produce a crosslinked foam fluid. The foam quality was determined by following equation:

$$\text{Foam}\% = \frac{V_f - V_s}{V_f} \tag{1}$$

Where V_s is volume of prepared solution; V_f is volume of prepared foam fluid.

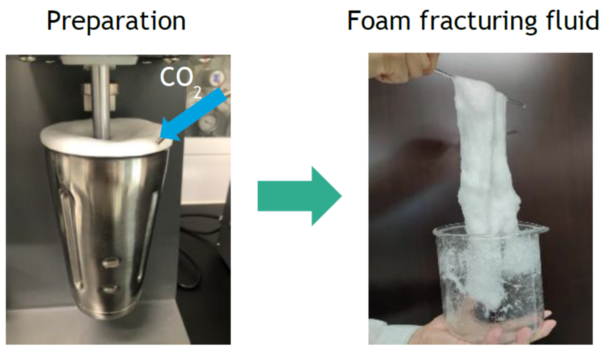

Figure 2—CO_2 foam fluid preparation

Stability evaluation of CO_2 foam fluid preparation

The prepared foam solution was promptly transferred into a sealed volumetric cylinder, which was then put into oven at measured temperature. The volumes of foam and liquid were monitored and recorded over time respectively. The half-life of foam was used to evaluate stability of CO_2 foam fluid.

Rheological measurement

Rheological properties of fluids were measured by Discovery HR-2 rheometer (TA Instrument). The correlation between apparent viscosity and shear rate was determined by varying shear rate from 0.01 s^{-1} to 500 s^{-1}. To assess thermal sensitivity, the apparent viscosity of the fluid was measured across a temperature range of 25°C to 150°C at 100 s^{-1}. Shear stability tests were conducted by maintaining a shear rate of 100 s^{-1} for 120 mins with temperature ranged from 90°C to 150°C.

Proppant suspending capacity measurement

The proppant suspending capacity of fluid was measured by static column test. A 200 ml prepared solution was uniformly mixed with 60 g ceramic proppants (size: 20~40 mesh, density: 1.56 g/cm^3) by high-speed mixer. The settling behavior of the proppants was observed, and the settling time was recorded to calculate the setting velocity.

Static break measurement

Prepared fluid was mixed with breaker and crosslinker in sequence. The obtained crosslinked fluid was then sealed and placed into oven with setting temperature for 12 hrs. After completion of the static aging period, the viscosity of the residual fluid was was measured to assess the effectiveness of the breaking process.

Coreflooding Evaluation for Retained Permeability

Coreflooding experiment was conducted to assess the retained permeability in the Berea sandstone coreplug following injection with filtered broken fracturing fluid. The broken fracturing fluid was filtered by Whatman membrane with pore size at 8 μm under 30 psi. The coreflooding experiment was conducted as per the following procedure:

1. Core plug samples were evacuate to remove air and saturated with 2 wt% KCl brine.
2. The saturated core was set in a core holder.
3. A confining pressure of 1000 psi and a backpressure of 500 psi were applied.
4. The oven temperature was then raised to 150 °C to simulate reservoir conditions.
5. 2 wt% KCl was pumped in production direction at varying flow rates to determine baseline differential pressure $\Delta P_{initial}$.
6. Approximately 3 pore volumes (PV) of the filtered, broken fracturing fluid were injected in the opposite (injection) direction with a constant rate of 1 mL/min.
7. The core was shut in for 2 hours to ensure the interaction between rock and fluid.
8. 2 wt% KCl was pumped again in production direction to determine post-treatment differential pressure ΔP_{final}.

The retained permeability was calculated using the follow equation:

$$\text{Retained Permeability}(\%) = \frac{\Delta P_{\text{initial}}}{\Delta P_{\text{final}}} \times 100\%$$

Where $\Delta P_{initial}$ is baseline differential pressure before treatment; ΔP_{final} is post-treatment differential pressure after treatment

Results

Chemical Screening and Evaluation

Foaming performance evaluation of surfactants. 5 surfactants were evaluated to explore their foaming potential with CO_2 in tap water. Table 3 shows the foam stability of CO_2 foams prepared using 500 ppm of surfactant in tape water. Surf-2 demonstrates comparable foaming performance to the benchmark Surf-5. Surf-3 and Surf-4 exhibit poor foaming capacity in tap water. Surf-1 is recommended as the preferred foaming agent for further development, due to the excellent foaming capacity and stability among the surfactants.

Table 3—Foaming Performance in Tap Water

Sample	Foam volume (ml)	Half-life (s)
Surf-1	600	240
Surf-2	350	87
Surf-3	40	10
Surf-4	0	0
Surf-5	360	86

Gelation performance evaluation of polymers with crosslinkers. Table 4 presents gelation performances of 12 commercial associative polymers with 2 organic zirconium crosslinkers. Both the polymer and crosslinker were used at concentrations of 0.5%. The objective of these experiments was to identify polymer-crosslinker combinations that are capable to form crosslinked gel within 5 mins. Among the tested formulations, only the combination of APAM-12 with OCX-1 can meet the rapid gelation criteria, proving its suitability for fracturing fluid.

Table 4—Gelation time of polymers with crosslinkers

Sample	OCX-1	OCX-2
APAM-1	Gelation in 30 mins	Gelation in 30 mins
APAM-2	Gelation in 2 hrs	Gelation in 30 mins
APAM-3	\	\
APAM-4	\	Gelation in 30 mins
APAM-5	\	\
APAM-6	\	Gelation in 30 mins
APAM-7	\	Gelation in 30 mins
APAM-8	\	Gelation in 30 mins
APAM-9	\	Gelation in 30 mins
APAM-10	\	\
APAM-11	\	Gelation in 30 mins
APAM-12	Gelation in 5 mins	Gelation in 30 mins

Note: "\" indicates no gel formed.

Development of Polymer-Based Fracturing Fluid

Thermal Impact on Viscosity Performance of Polymer Gels. Fig 3 presents the viscosity behavior of SCAP gels with and without the incorporation of Surf-1 across temperature ranged from 25°C to 150°C. The polymer and crosslinker concentrations were maintained at 0.5%. Polymer gel without Surf-1 exhibits typical thermal thinning behavior that is characterized by a steady decline in viscosity as temperature increases. In contrast, SCAP gel presents a distinct thermal thickening behavior with Surf-1 addition more than 0.2%, which is attributed to improved intermolecular interaction between surfactant and polymer. The viscosity of SCAP gels increases significantly, reaching a peak before gradually decreasing beyond 70°C. This decline is due to the diminished interaction by accelerated molecular movements with temperature raising. Notably, the SCAP gel can maintain a considerably higher viscosity than the one without Surf-1 addition, confirming that the presence of Surf-1 can remarkably reinforce intermolecular interaction with polymer molecules, resulting in increased viscosity and improved temperature tolerance. These findings highlight the potential of SCAP gel to deliver improved temperature tolerance, providing benefits for high temperature operation.

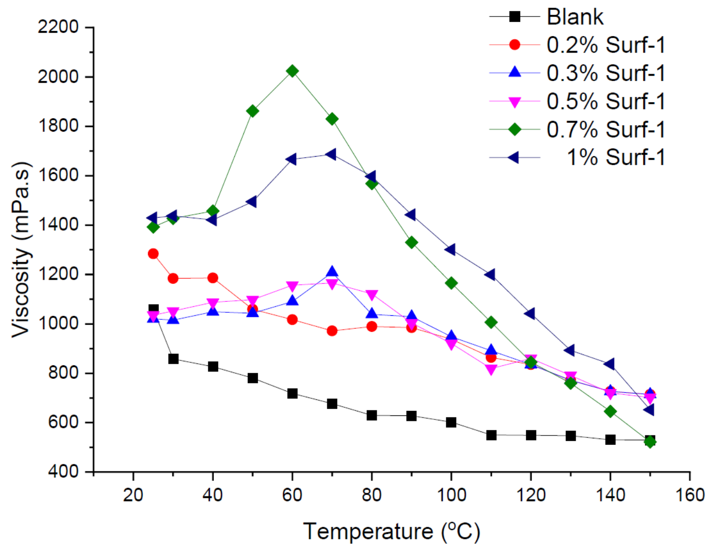

Figure 3—Effect of temperature on polymer gel fluids viscosity

Influence of Surfactant Addition on Shear Stability of SCAP Gels at High Temperature. SCAP gels formulated with different additions of surfactant Surf-1 were subjected to shear testing at 150°C for 2 hours under a constant shear rate of 100 s^{-1}. Polymer and crosslinker concentrations were consistently maintained at 0.5%. As presented in Fig. 4 SCAP gels exhibit good shearing stability with Surf-1 concentrations ranging from 0.2% to 0.5%, preserving retention viscosity above 300 mPa·s under high temperature. However, further increasing Surf-1 concentration beyond 0.5% leads to retention viscosity below the 300 mPa·s threshold, indicating reduced structural stability. According to these results, an optimal Surf-1 concentration of 0.5% is recommended to ensure good shear resistance and maintain high viscosity retention at 150°C.

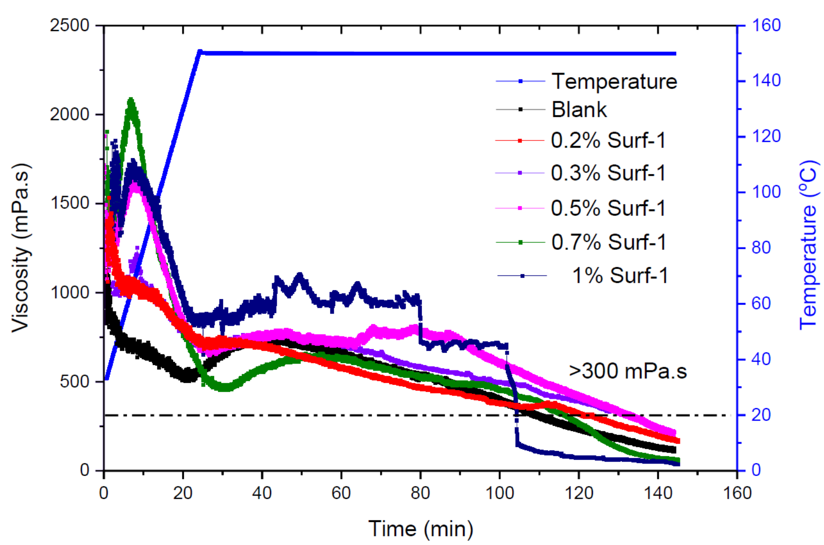

Figure 4—Viscosities of polymer gels with varying concentrations of Surf-1 as a function of time with shear rate of 100 s^{-1} at 150°C

Development of CO_2 foam fluid

Temperature Impact on CO_2 Foam Fluids Viscosity. Fig. 5 presents viscosity of CO_2 foam fluids with different foam qualities as a function of temperature. With temperature raising, viscosity reduction of foam fluids is observed. Besides, higher foam quality correlates with reduced viscosity levels. It is interesting to note that the viscosity variation becomes less pronounced at elevated CO_2 foam quality. This indicates the improved thermal tolerance of CO_2 foam fluid with increasing CO_2 foam quality, presenting promising application potential in thermally demanding environments.

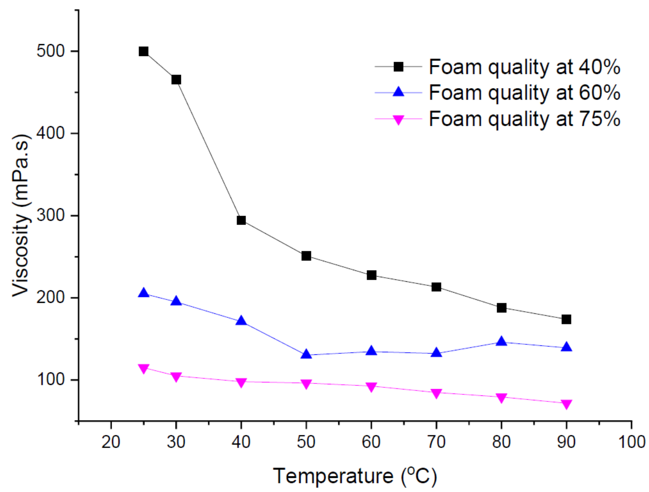

Figure 5—CO_2 Foam viscosity as a function of temperature with different foam qualities

Shear Resistance Performance of CO_2 Foam Fluids with Various Foam Qualities. The prepared CO_2 foam fluids with different foam qualities were sheared over 2 hours under 100 s^{-1} at temperature of 90°C. As shown in Fig.6, all the CO_2 foam fluids present relatively low viscosity ranging from 50 mPa.s to 150 mPa.s. Despite the low viscosity, the fluid presents good shearing stability especially for the sample with elevated foam quality, proved by the slight viscosity variations. Notably, the formulation with a foam quality of 60% % displays excellent shear resistance, with minimal viscosity loss and the highest retained viscosity of approximately 130 mPa·s among all samples. Based on these results, the recommended CO_2 foam quality for the developed formulation is around 60%.

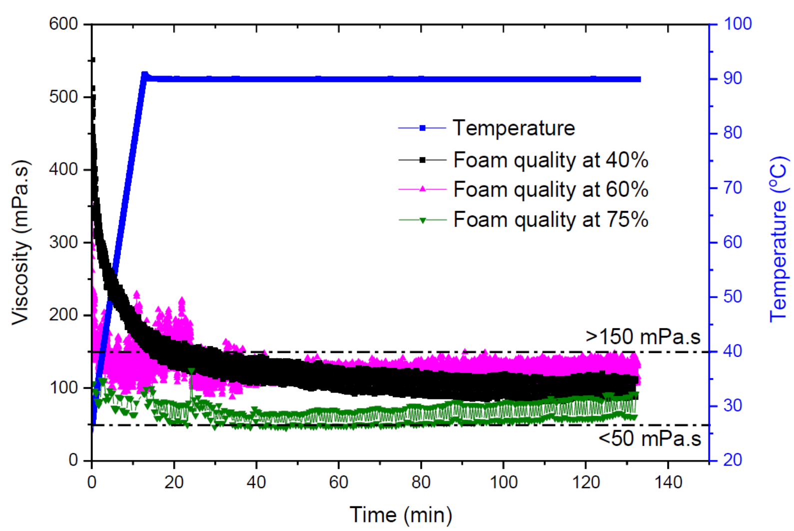

Figure 6—Viscosity of CO2 foam as a function of time with shear rate of 100 s^{-1} at 90°C

CO_2 Foam Fluids Performance with Different Polymer Concentrations. The prepared CO_2 foam fluids with different polymer additions were evaluated for shear stability at 90°C in 2 hours under a constant shear rate of 100 s^{-1}. The crosslinker and surfactant concentrations were held constant at 0.5% each. Fig. 7 indicates increasing polymer concentration can effectively improve foam fluid viscosity. CO_2 foam fluid with 0.8% polymer can remain viscosity above 300 mPa.s after the test, presenting good shearing stability under high temperature conditions.

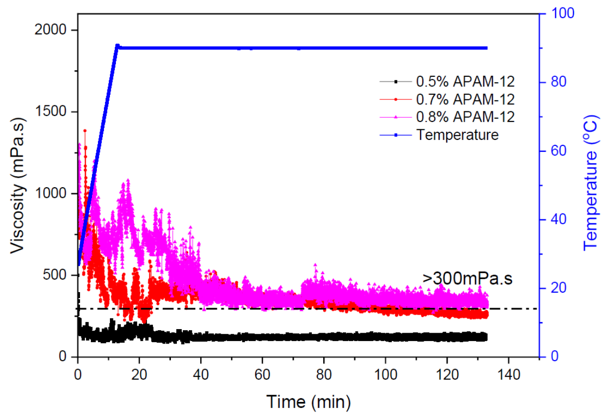

Figure 7—Viscosity of CO_2 foam fluid with different polymer concentrations as a function of time with shear rate of 100 s^{-1} at 90°C

CO2 Foam Fluids Performance under Different Temperatures. To evaluate the rheological performance of CO_2 foam fluids at elevated temperatures, CO_2 foam fluids were subjected to continuous shear at a rate of 100 s^{-1} for 2 hours at temperatures of 90°C, 120°C, and 150°C respectively. The tested formulation included 0.5% Surf-1, 0.8% APAM-12 and 0.5% OCX-1. Fig. 8 shows the developed foam fluid displays good shearing stability with temperature ranged from 90°C to 120°C. However, it presents unsatisfied performance at 150°C, limiting its application in higher temperature conditions.

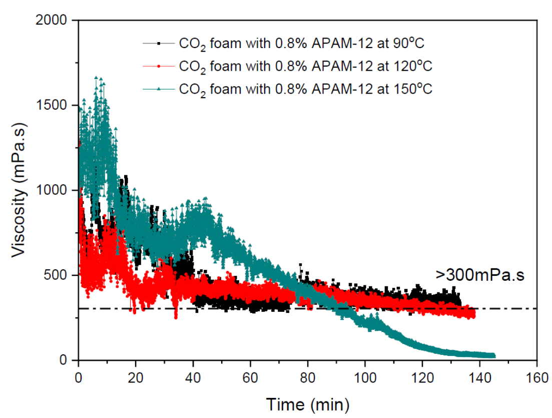

Figure 8—Viscosity of CO_2 foam fluid as a function of time with shear rate of 100 s^{-1} at different temperatures.

Enhancing CO_2 Foam Fluid Performance at 150°C with Thermal Stabilizer Additives. To address the performance limitation of CO_2 foam fluids at150°C, thermal stabilizer TU was introduced into the formulation. The foam fluids containing 0.5% Surf-1, 0.8% APAM-12, and 0.5% OCX were sheared at 100 s^{-1} for 2 hours with different additions of TU. Fig. 9 indicates TU can sharply improve foam fluid performance at high temperature. The foam fluids with TU addition ranged from 0.5% to 0.7% exhibits excellent performance at 150°C, retaining viscosity above than 300 mPa.s after the tests. These results proved the effectiveness of TU in reinforcing the thermal stability of foam fluids at extremely high temperatures, and the optimal TU concentration of 0.5% is recommended.

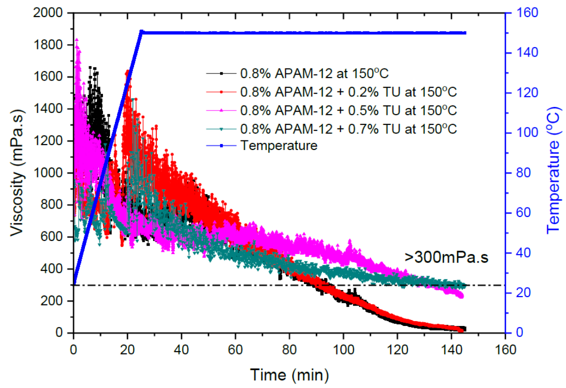

Figure 9—Viscosity of CO_2 foam fluid as a function of time with shear rate of 100 s^{-1} at 150°C.

Performance of The Developed CO_2 foam fluid package at 150°C

A CO_2 foam fluid package was further developed with additives including with clay stabilizer, flowback enhancer and breakers. The optimized formulation is shown in Table 5.

Table 5—The optimized formulation for CO_2 foam fracturing fluid

No.	Component	Chemical Name	Addition
1	Foaming agent	Surf-1	0.5%
2	Viscosifier	APAM-12	0.9%
3	Crosslinker	OCX-1	0.5%
4	Thermal stabilizer	TU	0.5%
5	Clay stabilizer	ClayS-1	0.1%
6	Flowback enhancer	Surf-5	0.1%
7	Breaker	B-Cap	0.02%
8	Gas	CO_2	60%

Stability Assessment of CO_2 Foam Fluid Package at 150°C. Foam stability evaluation of the developed package was conducted at 150°C to assess its performance at high temperature. Fig. 10 presents images of the foam before and after 3 hours test period. The visual comparison shows there is no obvious change due to foam maintaining the same volume, proving its excellent thermal stability at 150°C.

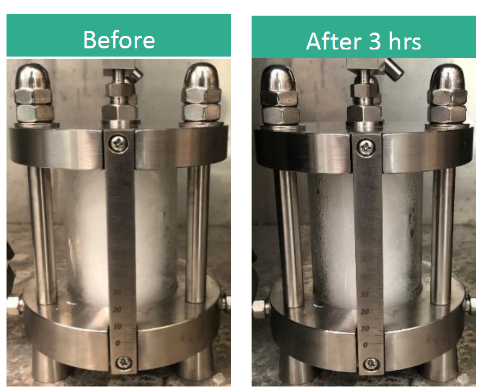

Figure 10—CO_2 Foam stability evaluation at 150°C

Proppant Suspension Performance of CO_2 Foam Fluids at 150°C. The proppant suspension capacity of CO_2 foam fluid prepared using the developed additive package with foam quality around 60% was investigated at 150°C. The pictures in Fig. 11 show there was little change before and after proppant suspension measurement, indicating good proppant suspension capacity at high temperature. In addition, the fluid volume exhibits slight variation over 3 hours which further supports the foam stability test results.

Figure 11—Proppant suspension performance at 150°C: Proppant/CO_2 foam: 30%

Rheological Performance of CO_2 foam fluid package at 150°C. Fig. 12 shows the rheological performance of the developed CO_2 foam fluid package under high temperature. The experiment was conducted at 150°C with shearing rate of 100 s^{-1} for 2 h. The CO_2 foam fluid maintained a viscosity of approximately 400 mPa·s after the test, demonstrating excellent performance for fracturing operation at high temperature conditions.

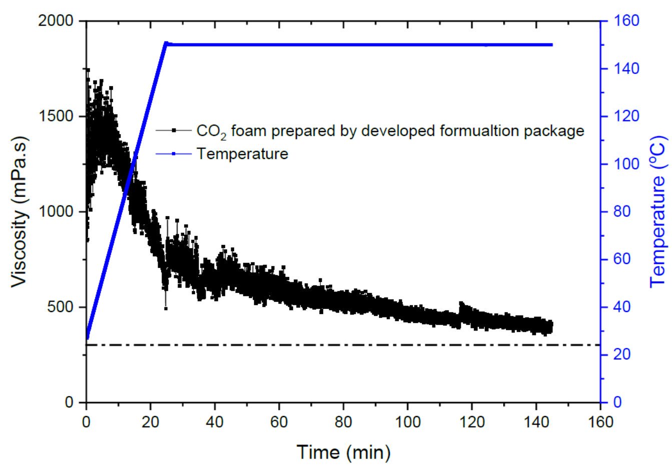

Figure 12—Viscosity of CO_2 foam fluid prepared in Example 1 as a function of time with shear rate of 100 s^{-1} at 150°C

Static Break Evaluation of CO_2 Foam Fluid Package. Residual fluids with high viscosity within the fracture can cause permeability reduction, diminishing the over all effectiveness of fracturing treatment. To address this, gel breakers are typically employed to significantly degrade polymer into small molecular weight fragment to achieve a sharp viscosity reduction and enable sandpack rapidly clean up. A capsule breaker B-Cap was selected for the developed foam fluid package. Static break tests for foam fluids with 0.02% of capsule breaker was conducted at 150°C for 12 hr and the results are presented in Fig. 13. It demonstrates CO_2 foam with 0.02% break undergoes significant degradation, altering into a watery fluid with viscosity less than 5 mPa.s at high temperature after break test, indicating its low formation damage.

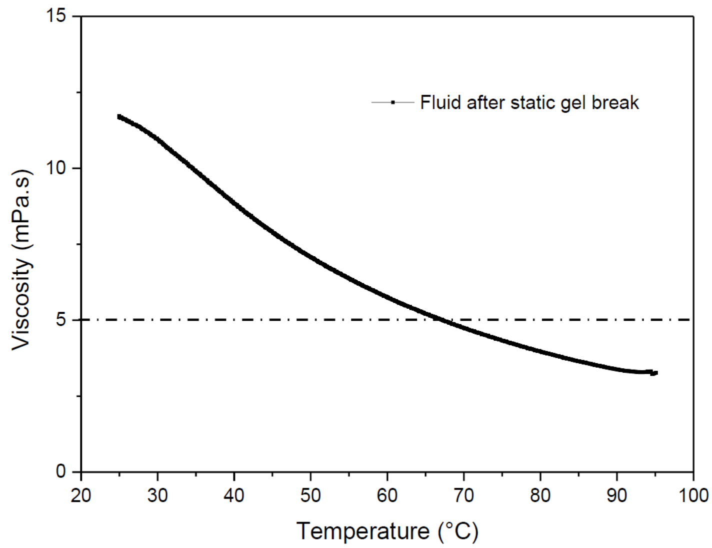

Figure 13—Viscosity of CO_2 foam fluid after break as a function of temperature with shear rate of 100 s^{-1}

Retained Permeability Assessment of CO_2 Foam Fluids after Static Degradation. The formation damage associated with the developed CO_2 foam fluid package was evaluated by core flooding experiments. The retained permeability of a Berea sandstone core plug was assessed following treatment of the broken CO_2 foam fluid. A Berea sandstone core plug with permeability around 80.23 mD was selected for the assessment (Table 6). The testing temperature was 150°C. During the injection of the filtrated fracturing fluid, the differential pressure across core plug increased from 0.28 psi to 1.45 psi approximately (Fig. 14). When 2% KCl brine was re-injected through opposite direction across core plug, the pressure drop decreased to around 0.32 psi. The resulting retained permeability is nearly 87.5% indicating excellent fracturing fluid clean-up.

Table 6—Core plug information

Core No.	Diameter (cm)	Length (cm)	Dry weight (g)	Wet weight after brine perm. (g)	Pore volume (ml)	Brine perm. (md)
38A	3.8	5	123.8879	135.8540	11.9661	80.23

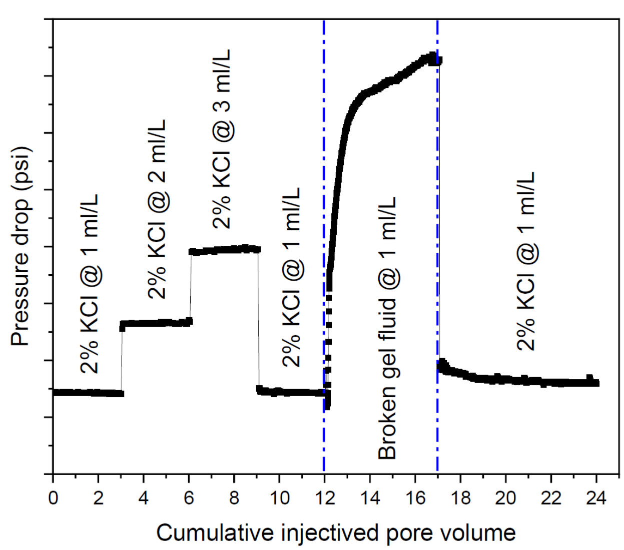

Figure 14—Retained permeability of developed CO_2 foam fracturing fluid

Conclusions

1. A comprehensive screening of 5 surfactants, 12 commercial associative polymers and 2 organic zirconium crosslinker were conducted to investigate foaming capacity and gelation performance. A surfactant crosslinked associated polymer formulation (SCAP) was developed using screened chemicals. The addition of Surf-1 can effectively improve fluid viscosity and temperature tolerance, resulting from the intermolecular interaction between surfactant and associative polymer. The fluid exhibits thermal behavior within temperatures less than 70°C compared with the one without Surf-1. The optimized SCAP gel fluid including 0.5% Surf-1, 0.5% APAM-12 and 0.5% OCX-1 presents good performances at high temperature conditions, which can preserve viscosity above 300 mPa.s at shear rate of 100 s^{-1} over 2 hours with temperature up to 150°C.
2. A CO_2 foam fracturing fluid was developed upon the SCAP gel formulation. CO_2 foam with high foam quality presents good temperature tolerance and shearing stability. Increasing polymer concentration proves effective in achieving the target viscosity of 300 mPa·s. Addition of thermal stabilizer TU can enlarge application of CO_2 foam fluid at 150°C due to the sharply enhanced thermal stability. The optimized CO_2 foam fracture fluid formulation comprising 0.5% Surf-1, 0.8% APAM-12, 0.5% OCX-1, and 0.5% TU with 60% foam quality can maintain high viscosity above 300 mPa.s at shear rate of 100 s^{-1} over 2 hours with temperature up to150°C.
3. A CO_2 foam fracturing fluid package was further finalized, consisting of 0.5% Surf-1, 0.9% APAM-12, 0.5% OCX, 0.5% TU, 0.1% ClayS-1, 0.1% Surf-5 and 0.02% B-Cap. The optimized formulation presents good performances in foam stability, proppant suspending capacity and high retention viscosity at 150°C. Static break tests confirm complete degradation of CO_2 foam into a watery fluid, meeting the field operational criteria. The coreflooding test further validated low formation damage, with the retained permeability reaching around 87.5%.

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